



BGC TRUST UNIVERSITY BANGLADESH

Department of Computer Science & Engineering

Assignment 01

Course Title : Software Development Project - 1

Course Code : 07-0613-SDP509

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Submitted To

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1. Project Title

University Academic Archive & Management System

“A centralized platform for academic record management, thesis archiving, and seamless communication between students and professors.”

2. Description of the Project

The **University Academic Archive & Management System** is an **all-in-one digital platform** designed to automate and centralize academic operations in universities.

Key Features:

- **Student Portal:** Students can track attendance, view CT marks, assignment marks, exam results, submit projects, and communicate with professors.
- **Professor Portal:** Professors can update academic records, approve/reject projects, respond to student messages, and monitor performance.
- **Secure Archive:** All student projects and thesis are stored digitally for easy access and future reference.
- **Analytics:** Tracks downloads, top-performing students, and department-wise performance trends.

This system combines academic management, communication, and archive in **one unified platform**, reducing administrative burden and improving efficiency.

3. Motivation

Universities often manage academic records manually, which is:

- **Time-consuming**
- **Prone to errors**
- **Inefficient for thesis/project tracking**

The motivation behind this project is to **simplify academic operations, digitize thesis/project archives, and streamline communication** between students and professors. It ensures **accuracy, transparency, and efficiency** in academic management.

4. Objectives

1. Digitally centralize student academic records (attendance, marks, projects).
2. Provide a secure repository for thesis and projects.
3. Enable smooth communication between students and professors.
4. Facilitate performance tracking and analytics.
5. Reduce workload for university staff and faculty.
6. Build a scalable system that can be enhanced in the future.

5. Contribution

- **Students:** Easy access to academic records, project submission, and communication.
- **Professors:** Efficient management of attendance, marks, and project approvals.
- **University:** Centralized archive, secure storage, and analytics for performance monitoring.
- **Overall:** Reduces paperwork, improves transparency, and streamlines operations.

6. Structured Approach or object-Oriented Approach

The project is developed using **Object-Oriented Programming (OOP)** principles.

Key Classes and Responsibilities:

Class	Attributes	Methods
Student	student_id, name, email, department	submitProject(), viewMarks(), sendMessage()
Professor	professor_id, name, email, department	approveProject(), updateMarks(), replyMessage()
Project	project_id, title, abstract, file_path, student_id, supervisor_id, status	upload(), approve(), download()
Attendance	attendance_id, student_id, course_id, total_classes, attended	markAttendance(), viewAttendance()
Marks	mark_id, student_id, course_id, ct_marks, assignment_marks, exam_marks, total	updateMarks(), calculateTotal()
Message	message_id, sender_id, receiver_id, message, timestamp	send(), reply(), markRead()

OOP Advantage: Modular, maintainable, and scalable for future updates.

7. SDLC Model

We adopt the **Waterfall Model** for structured development:

1. **Requirement Analysis:** Gather requirements from students, professors, and admin.
2. **System Design:** Database, class design, user interface mockups.
3. **Implementation:** Develop frontend and backend modules.
4. **Testing:** Unit testing, integration testing, and system testing.
5. **Deployment:** Deploy system for university use.
6. **Maintenance:** Bug fixes and feature enhancements.

8. Gantt Chart (6 Month)

Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Requirement Gathering & Analysis						
System & Database Design						
Class Structure & Backend Setup						
Frontend Design & UX/UI						
Module Integration						
Testing & Debugging						
Deployment & Documentation						

9. Functional and Nonfunctional Requirements

Functional Requirements:

- Student login and dashboard
- Professor login and dashboard
- Attendance tracking
- CT, assignment, and exam marks management
- Thesis/project submission, approval, and download
- Messaging between students and professors
- Analytics dashboard for admin

Nonfunctional Requirements:

- Secure login authentication
- Reliable and scalable database
- Responsive and intuitive UI
- Fast data retrieval and processing
- Data backup and recovery mechanisms

10. System Architecture

Three-tier architecture:

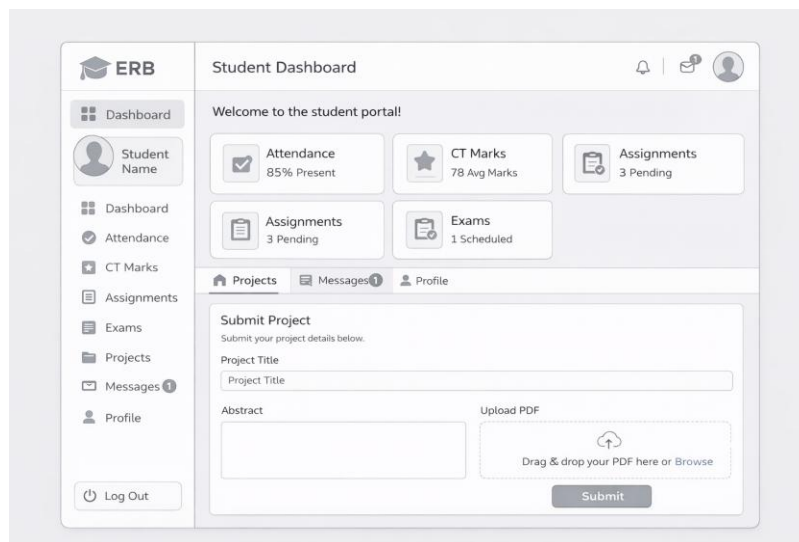
1. **Presentation Layer (Frontend):** Student & professor portals, web pages or GUI.
2. **Business Logic Layer (Backend):** Handles operations like marks calculation, project approval, messaging.
3. **Data Layer (Database):** Stores students, professors, departments, marks, attendance, projects, messages.

[Student/Professor Portal] ---> [Backend Server/API] ---> [Database Server]

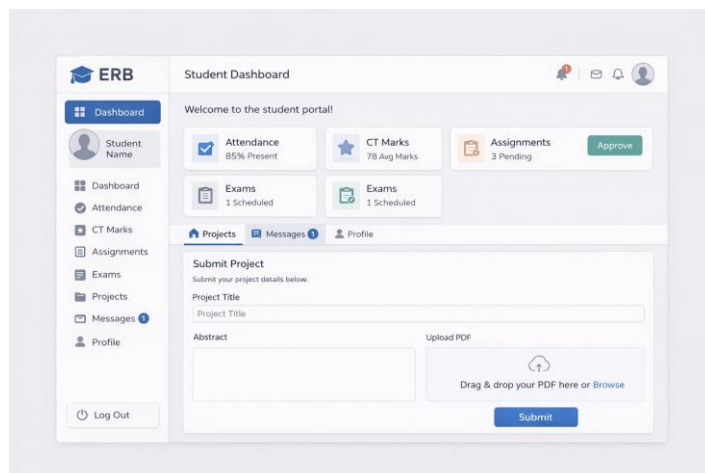
11. UX/UI Design

Student Portal Wireframe:

- Dashboard: Attendance, Marks, Projects, Messages
- Project Submission Page: Upload PDF, Title, Abstract
- Message Page: Inbox / Send Message

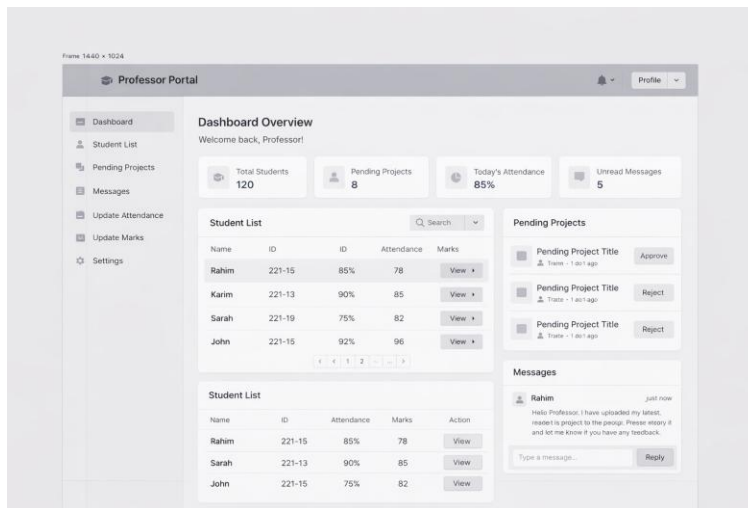


UI Design:

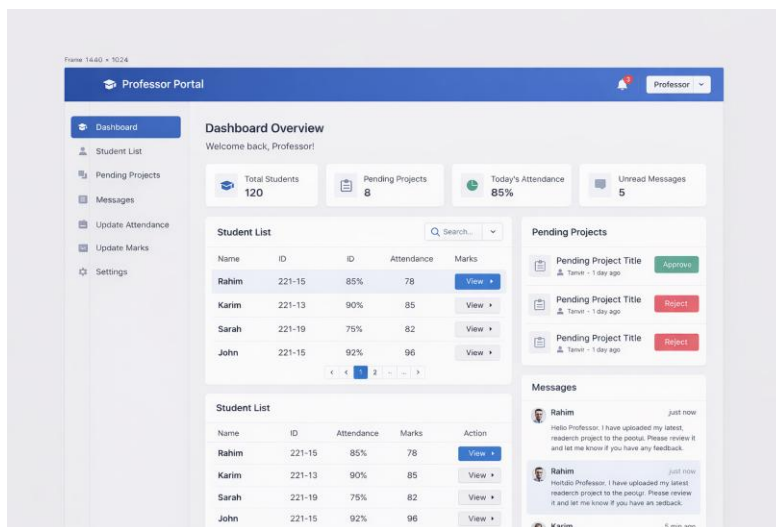


Professor Portal Wireframe:

- Dashboard: Student list, pending project approvals
- Update Marks / Attendance Page
- Message Inbox



UI Design:



12. Frontend Languages

- HTML, CSS, JavaScript
- react js (Optional)

13. Backend & Database Languages

- Backend: PHP
- Database: MySQL